

ElementReview and JudgeVideoReplay Installation and Setup Manual

This manual covers a normal event setup with one VRO computer and separate judge/referee computers.

Which App Goes Where

Install `ElementReview` only on the VRO computer.

The VRO computer records the incoming video, creates the replay files, runs the local ElementReview server, and serves replay clips to the judging panel.

Install `JudgeVideoReplay` on every judge and referee computer.

JudgeVideoReplay is the panel replay client. It does not record video and does not replace ElementReview on the VRO computer.

Before Installing

Confirm that all computers are on the same event network and that the VRO computer has a stable (not dynamic) IP address. The judge and referee computers will use that VRO IP address to connect to ElementReview.

On Windows 10 computers, ElementReview and JudgeVideoReplay require the Microsoft Edge WebView2 Runtime. If either app opens to a blank window or fails immediately on a Windows 10 computer, install the [Microsoft Edge WebView2 Evergreen Runtime](#) and start the app again. Do not install WebView2 separately on Windows 11 as this runtime is already included in Windows 11.

Recommended Computer Requirements

ElementReview does the heavier work. It records the incoming stream, creates high-res and low-res replay video, serves replay files to the panel, and writes/deletes a large amount of video data during an event.

Minimum recommended VRO computer for ElementReview:

Component	Minimum recommendation	Preferred event setup
CPU	Recent Intel Core i5/i7 or AMD Ryzen 5/7, 4 cores or better	Intel Core i7/i9 or AMD Ryzen 7/9, 6 cores or better
RAM	16 GB	32 GB if available
Graphics	Integrated graphics may work for basic use	Hardware video encoding support is strongly preferred, such as Intel Quick Sync, NVIDIA NVENC, or AMD hardware encoding
Storage	SSD, 1 TB minimum with plenty of free space	
Network	Wired Gigabit Ethernet	

Do not use a mechanical hard drive for the VRO computer. ElementReview writes and deletes video data continuously, so an SSD gives better recording/replay performance. A 1 TB or larger SSD is recommended because smaller drives can fill quickly and may wear faster when used repeatedly for video recording workloads.

For JudgeVideoReplay computers, the requirements are lighter because they receive and play back low-res replay video rather than recording and encoding the main feed.

Minimum recommended judge/referee computer:

Component	Minimum recommendation
CPU	Recent Intel Core i3/i5 or AMD Ryzen 3/5
RAM	8 GB
Graphics	Standard integrated graphics are usually sufficient
Storage	SSD recommended
Network	Reliable Wi-Fi or Ethernet

The VRO computer must use wired Gigabit Ethernet. Judge and referee laptops can usually run successfully on a good closed event Wi-Fi LAN, provided the wireless network is not shared with spectators or general venue traffic.

Windows Defender and SmartScreen

Windows may warn about newly downloaded or newly provided installers, especially before the app has a broad reputation with Microsoft SmartScreen.

If Windows SmartScreen blocks the installer:

1. Click **More info**.
2. Confirm the publisher/file is the expected ElementReview or JudgeVideoReplay installer.

3. Click **Run anyway**.

If Windows Defender or the browser marks the file as suspicious:

1. Confirm the file came from a trusted source.
2. If Windows shows a **Keep** or **Keep anyway** option, choose it only after confirming the file source.
3. If the file is blocked after copying, right-click the installer, choose **Properties**, check **Unblock** if present, then click **OK**.

Do not bypass these warnings for files from an unknown source.

Install ElementReview on the VRO Computer

1. Run the **ElementReview-Setup-<version>.exe** installer on the VRO computer.
2. Start **ElementReview**.
3. Open the settings screen by clicking the gear icon located at the top right of the window.
4. Configure the CSS link, video source, encoding, and saved-video settings.
5. Save settings and restart ElementReview.

Install JudgeVideoReplay on Judge and Referee Computers

1. Run the **JudgeVideoReplay-Setup-<version>.exe** installer on every judge and referee computer.
2. Start **JudgeVideoReplay**.
3. Open the settings screen by clicking the gear icon located at the top right of the window.
4. Set **Server IP address** to the IP address of the VRO computer.
5. Set the role to **Judge** or **Referee**.
6. Save settings.

The **Server IP address** must be the VRO computer IP address, not the judge/referee computer IP address.

Example:

```
192.168.6.60
```

ElementReview Initial Configuration

Open ElementReview settings on the VRO computer by clicking the gear icon located at the top right of the window.

CSS Integration

If integrating ElementReview with Legacy CSS, set **CSS Link Type** to:

```
Legacy CSS
```

Then, set **MSSQL Database Host** to the IP address of the CSS database host (normally the EC computer).

If integrating ElementReview with Online CSS or Offline CSS, set **CSS Link Type** to:

None

The display of executed element codes and automatic determination of discipline/category/segment-specific halfway time values is currently only supported for Legacy CSS. Similar support for Online CSS and Offline CSS is forthcoming.

When CSS integration is set to **None**, ElementReview cannot automatically detect category, discipline, segment, or halfway timing from CSS data. Use the manual **HW:** dropdown in the recording/replay UI to manually select the appropriate halfway time value when halfway timing is needed.

Video Source

Turn on Demo Mode for training or demonstration purposes. This mode uses a locally stored video instead of an RTSP video stream as an input source. The local video can be customized by replacing the file demovideo.mp4 located in the User\AppData\Local\ElementReview\data folder.

Supported video format:

Container: MP4
Video codec: H.264 / AVC
Frame rate: constant 30 or 60 fps
Resolution: 1920x1080 preferred, 1280x720 acceptable
Audio: optional; not needed for Demo Mode
Fast start: enabled
Avoid: HEVC/H.265, HDR, variable frame rate, unusual codecs

Turn off Demo Mode for real event recording.

Set the RTSP URL to the video encoder stream URL.

Example without a special port:

```
rtsp://192.168.6.200/0
```

Example with an explicit port and path:

```
rtsp://192.168.1.168:8554/video
```

Set **RTSP Transport Protocol** to **UDP** in most cases. UDP is usually the best choice on a clean, local event network because it has lower latency.

Use **TCP** only if the network is unreliable or if instructed by Skate Canada technical support.

Encoding Settings

Recommended settings:

Setting	Recommended value	Notes
High-res Video GOP	10	Used by ElementReview on the VRO computer. Lower GOP helps responsive seeking.
Low-res Video GOP	30	Used by JudgeVideoReplay clients.
Low-res Video Bitrate	2500 to 4000 kbps	Higher values improve quality but use more network bandwidth.

Use 2500 kbps when bandwidth is limited or many clients are connected. Use 4000 kbps when the event network is strong and better judge replay quality is desired.

Enable **Use Hardware Encoding** in most situations. Disable it only if directed by Skate Canada technical support or if hardware encoding causes a confirmed issue on the VRO computer.

Firewall and Network Requirements

The VRO computer must allow judge and referee computers to connect to ElementReview.

Required inbound rule on the VRO computer:

TCP 5050

ElementReview listens on:

http://0.0.0.0:5050

JudgeVideoReplay uses TCP port 5050 for status, replay data, and video file downloads from the VRO computer.

The code also uses local sidecar services on the VRO computer, including:

TCP 8889
 UDP 8189
 TCP 8554 on 127.0.0.1
 TCP 9997 on 127.0.0.1

These sidecar ports are used by ElementReview and MediaMTX on the VRO computer. JudgeVideoReplay clients do not normally need direct access to them. The important client-facing firewall requirement is inbound TCP 5050 on the VRO computer.

If Windows Firewall prompts when ElementReview first runs, allow access on the event/private network.

If connections still fail, options include:

- Add an inbound Windows Firewall rule allowing TCP 5050 for **ElementReview.exe**.

- Add an inbound Windows Firewall rule allowing TCP 5050 for the event network profile.
- Temporarily turn off Windows Firewall on the VRO computer for the event network.

Turning off Windows Firewall is simple for troubleshooting, but a targeted TCP 5050 allow rule is preferred when possible to maintain network security.

Connection Checklist

On the VRO computer:

1. Start ElementReview.
2. Confirm the VRO computer is connected to the event network.
3. Confirm the VRO IP address.
4. Confirm TCP 5050 is allowed through the firewall.

On each judge/referee computer:

1. Start JudgeVideoReplay.
2. Open the settings screen by clicking the gear icon located at the top right of the window.
3. Enter the VRO computer IP address.
4. Select the correct role (Judge or Referee)
5. Save settings.
6. Confirm JudgeVideoReplay connects when ElementReview is running.

Troubleshooting

If JudgeVideoReplay stays on the waiting screen:

- Confirm ElementReview is running on the VRO computer.
- Confirm the VRO IP address in JudgeVideoReplay is correct.
- Confirm all computers are on the same LAN/VLAN.
- Confirm the event network does not block client-to-client traffic.
- Confirm TCP 5050 is open inbound on the VRO computer.

If ElementReview does not show video:

- Confirm Demo Mode is off for live event recording.
- Confirm the RTSP URL is correct.
- Confirm the encoder is powered on and connected to the network.
- Try UDP first, then try TCP if the stream is unstable or unavailable.
- Confirm the encoder IP address is reachable from the VRO computer.

If judge replay video is low quality or stutters:

- Increase low-res bitrate toward 4000 kbps (possibly higher) for better quality if bandwidth allows.
- Reduce low-res bitrate toward 2500 kbps (possibly lower) if the network is congested.
- Confirm all clients have strong wired or wireless network connectivity.
- Keep low-res GOP at 30 unless directed otherwise.